

# Multimodal EMI Instruction for Enhancing Primary Students' Engagement and Speaking Proficiency: A Quasi-Experimental Study

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Received: ....../...../2025

Revised: ....../...../2025

Accepted: ....../...../2025

## Abstract

In response to Vietnam's national initiative to promote English as a second language in schools by 2035, English-medium instruction (EMI) has been increasingly implemented in primary education to increase students' exposure to English through academic subjects. However, limited research has examined how to effectively teach abstract concepts while simultaneously enhancing young learners' oral communicative competence in EMI settings. This study aims to investigate the effectiveness of multimodal instruction integrating embodied modes and disembodied modes in improving students' engagement and speaking performance in Science classes. A quasi-experimental design was employed in a public school, involving an experimental group ( $N = 43$ ) and a control group ( $N = 40$ ). The 9-week intervention was developed using the ADDIE model and implemented through dynamic assessment and oral presentation tasks. Students' performance was evaluated using a speaking assessment rubric adapted from Harris (1974) and Brown (2004). Paired-samples and independent-samples t-tests revealed that students in the experimental group significantly outperformed those in the control group in terms of speaking performance. The findings suggest that the multimodal approach, particularly embodied modes, can effectively enhance student engagement and oral fluency in EMI contexts. This offers practical implications for designing multimodal EMI lessons in primary education. The study contributes to the limited body of research on multimodal approaches in primary EMI contexts and highlights the critical role of embodied learning in facilitating students' understanding of academic subjects, as well as the interplay between physical actions and cognitive processes in EMI learning contexts in Vietnam.

**Keywords:** *multimodal input, English-medium instruction, classroom interaction, speaking proficiency, EMI class*

## Introduction

In the context of Vietnam's national project to make English its second language in schools by 2035, many primary schools have officially implemented English-medium instruction (EMI) in everyday English, Math, and Science classes. EMI aims to increase students' exposure to English while enabling them to access global knowledge through academic subjects. This might also bring advantages in improving students' listening and speaking skills in English because of the increase in their exposure to English. Since then, relying solely on verbal language may not be sufficient to help them understand profoundly due to a number of barriers; most importantly, English is not their first language. Similarly, teachers, in the process of teaching subjects, might be prevented from aiming for the objectives of the lessons due to the difficulty of using verbal language only to convey the lessons. As a result, they might be preoccupied with explaining extra information for students or use inappropriate methods, teacher-centered approaches...to deliver the lessons, which might cause unexpected problems such as using advanced words or long, explanatory speech. Consequently, this might also reduce the students' interaction time and meaningful language use.

Different systematic studies have investigated the integration of diverse semiotic resources that serve as a natural extension of translanguaging, enabling students to engage in intricate processes of meaning-making (Zarobe & Querol-Julián, 2021). Some studies have focused on specific semiotic modes for meaning-making in classrooms—for example, gestures (Bezemer, 2014; Cao and Chen, 2017; Lim, 2017), classroom space (Lim et al., 2012), gestures and facial expressions

(Sueyoshi and Hardison, 2005), or language and images (Lotherington et al., 2019) which facilitate students' understanding in abstract contexts. In addition, in order to maximize learning, teachers often make great efforts to motivate students to participate in class interaction (Peng et al., 2017; Peng, 2019). Morell et al. (2020) uncover that trained English-medium instruction (EMI) lecturers combine semiotic resources (e.g., gaze, gesture, and written language) with their use of verbal discourse to engage students, and the orchestration of communicative modes serves to foster engagement in the classroom.

However, most existing studies have examined isolated semiotic resources such as gestures, gaze, facial expressions, or classroom space. Also, most of the research mentioned the role of multimodal resources in supporting meaning-making and classroom engagement, not paying much attention to how to improve speaking proficiency in EMI contexts. There also remains limited evidence regarding the effectiveness of integrated multimodal task designs. Hence, the research into multimodal pedagogies in promoting authentic language use in speaking and learner motivation in Vietnam needs more empirical studies to explore particularly hands-on learning activities combined with presentations in improving both learners' speaking performance and students' engagement. Also, only a few studies have shown that a teacher's facial expressions, gestures, and spatial positions are fundamental communicative modes that contribute to EFL learners' willingness to communicate in English language classrooms (Peng et al., 2017; Peng, 2019, Morell, 2018). Therefore, this study investigates the impact of multimodal task design, particularly hands-on learning activities integrated with student presentations in each lesson of one topic. By addressing these gaps, the study seeks to contribute to a deeper understanding of how multimodality can be effectively leveraged to enhance students' engagement and also students' speaking proficiency at primary schools in Hanoi. To meet the aim of the study, the following questions are put forward:

1. To what extent does the use of multimodal instruction improve students' speaking proficiency in the EMI class?
2. How do students respond to multimodal instruction in terms of classroom engagement?
3. How is multimodal instruction integrated in different stages of the lesson to support students' engagement and speaking proficiency?

## **Literature Review**

### **An overview of English as a Medium of Instruction**

Implementing EMI in a number of subjects at schools, particularly public schools, is essential to achieving the goal of using English as a second language in a variety of countries worldwide. Even though there are still issues with operation and implementation, using EMI would help students become more fluent in the language, expand their knowledge in other disciplines, and view English as a tool rather than a subject as it is in Vietnam. EMI Oxford, a group of researchers in the Department of Education at the University of Oxford has stated the following: "The use of the English language to teach academic subjects (other than English itself) in countries or jurisdictions in which the majority of the population's first language is not English." (Macaro, 2018). This definition is still tied to the author's uncertainty because the research context is not entirely credible because the United Kingdom is not an EMI country.

However, it would seem that EMI has spread almost everywhere. For now we can confidently report that EMI is a growing global phenomenon in all phases of education and educational settings (Dearden, 2015). This explicitly states that EMI is a tendency and a necessary implication in non-native speaking countries in the direction of actively making it a second language in diverse countries. Therefore, EMI means English is not only used as the content of the subject, but it is also the language of instruction for some subjects in schools. Furthermore, the more time students acquire languages in various contexts, the more fluent they could be in different English skills. The communicative approach to language teaching (CLT) postulates that it is essential for students to interact in a language frequently if they aspire to learn it (Holliday 1994). Therefore, it is necessary for students to expand their time using English, not only in English subjects at school, but also using English to explore other fields such as Maths and Science. The adoption of English as a medium of instruction seems to be anchored on best practices in ELT. The present study highlights the need of combining the aim of improving speaking proficiency and engagement in an EMI context in primary education, which was not broadly focused on by other researchers.

### **An overview of multimodal theory in Language Education**

In the context of Content and Language Integrated Learning (CLIL) and English Medium Instruction (EMI), multimodality can enhance both the content and the language learning objectives, as well as foster the development of critical thinking, creativity, and intercultural awareness (Lo & Lin, 2015; Lo & Leung, 2022). The study uses multimodal instructions to improve learning effectiveness for students and teachers working in primary education settings, addressing the issue of students' low English language proficiency. Multimodality seeks to examine and explain the way in which communication is built by using a series of semiotic resources that, when suitably integrated, make it possible to transfer meanings. Multimodality refers to the use of various semiotic modes alongside language to facilitate learning:

- embodied modes, such as gestures and facial expressions; and
- disembodied modes, such as visual aids and digital media (Norris, 2004); and how
- interpersonal relationships are established in EMI classrooms.

Within CLIL/ EMI classrooms, the role of multimodality has gained increasing recognition as a critical component of effective communication and learning (Liu & Lin, 2021). Except for the factors mentioned above, culture-specific extra-linguistic modes of expression, including corporeity, like deictic/metaphoric gestures, glances and visual orientations, posture, and physical space-positioning/movement (Kress et al., 2001), are considered useful tools to express emotions and to elaborate on verbal messages.

It is also vital for teachers to have a solid understanding of multimodal practices in order to implement them in the empirical education setting and produce a certain output after various lessons. In this sense, the need for EMI teachers to develop their multimodal competence, understood as the ability to exploit semiotic resources to construct meanings (Royce, 2002), appears to be evident. Whereas, multimodal literacy can be understood as the user's ability to navigate and produce texts that combine different semiotic resources (Jewitt & Kress, 2003). The research attempts to bridge the gap between the integration of language, content, and

multimodality in primary schools. This aims to increase students' engagement in the classroom while decreasing teachers' dominance regardless of teachers' English proficiency, as traditional teaching approaches in schools have not impacted students' language proficiency.

### **Assessing students' oral communicative competence in EMI contexts**

Most of the assessment activities used in both types of EMI programs reported in the dataset focused on assessment of learning (measuring and reporting the level of students' accomplishments—such as final exams) (L. T. T. Truong et al, 2020). These assessment types are considered instrumental and allow little room to motivate students' intrinsic learning desires (Earl & Katz, 2006). To change the way that students have access to the knowledge, the assessment for learning (integrated into teaching to enhance learning—such as oral presentations) should be a significant means of assessment throughout their EMI experience. (L. T. T. Truong et al, 2020). Dynamic Assessment (DA), a classroom-based assessment tool (Poehner and Wang, 2021), should be brought into this type of environment and, combining with the notion of the zone of proximal development (ZPD) (Vygotsky 1978: 86). This approach was used in the research to observe the needs of changing the aspects of multimodality insertion in the class. In addition, the researcher also combines the benchmark of speaking criteria, which Harris (1974) has stated has five primary components to assess learners' speaking performance. In the context of EMI classes, academic speakers are required to have more *micro*- and *macro*-skills of speaking (Brown, 2004:142-143). This mainly focuses on several key words such as "ability of producing chunks," "stress patterns," "adequate number of lexical units," "fluent speech at different rates," and so on.

Whereas other studies could not grasp and adjust the intervention during the research, combining two ways of assessing students' speaking performance allows teachers to be able to adjust the need of modifications in the process of intervening in different modes of modality approach. Since then, different aspects of speaking performance are assessed actively during and after the lesson to achieve fluency and content mastery.

## **Methodology**

### **Pedagogical Setting & Participants**

The research was conducted with the participation of 2 groups of teachers and their following students: (i) The control group (CG): This group of teachers and students was learning science in the class. They were from a public primary school with around 40 students in a class. (ii) The experimental group (EG): This group of teachers and students was learning science. They were from the same public school as the CG group, with around 43 students in a class. All participants were fourth-grade students from the same public primary school in Hanoi. All of those students and teachers had the same background of culture, politics, and social contexts when conducting this research. Their English proficiency was approximately at the A1 level according to the CEFR.

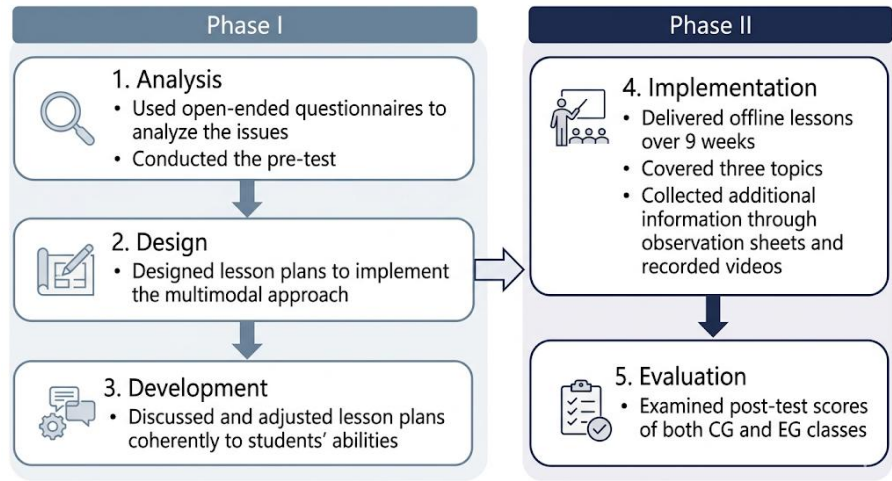
### **The reality in EMI classes**

To capture the fact of learning and teaching in EMI classes, the researcher collaborated with two teachers in the same school to evaluate several possible difficulties and the normal process of a daily EMI lesson using a Likert-5 scale. Both teachers used American Textbook Reading with 2 periods each week. The steps they used were quite the same, with three steps: presentation, practice, and production. Both of the teachers encountered the problems related to delivering abstract words in a simple way, and there were several hard technical terms in the limited time of 25 minutes. Their solutions mostly were to translate key words into Vietnamese to help them comprehend the lessons profoundly. However, the results of students’ speaking proficiency were relatively low in both environments, although they have made efforts in facilitating the words by using pictures in the textbook.

**Design of the Study**

A quasi-experimental research method was used to investigate the impact of multimodality on the learning experiences of students and teachers, as well as their engagement and speaking proficiency (Reichardt, 2009; Shadish et al., 2002). The research topic inherently involves both quantitative and qualitative aspects. The study lasted nine weeks and involved three meetings during the second semester of the 2024-2025 school year. The research design follows the generic ADDIE model (Peterson, 2003).

The design and development process are visualized in Figure 1.



**Figure 1: ADDIE flow diagram (Peterson, 2003)**

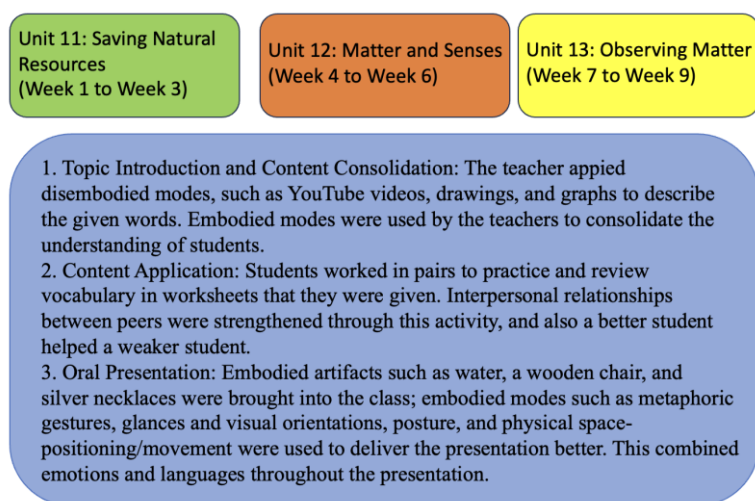
**Data collection & analysis**

The results of this study are based on both quantitative and qualitative data collected by CG and EG and by the researcher likewise. To diminish selection differences in a quasi-experimental design (Reichardt, 2009), a pre-test-posttest design was adopted. The quantitative data was used to collect science test scores, evaluated by a third-party teacher to ensure the transparency of the results. Notably, the examiners were asked to follow the scale 1-20 for five main speaking criteria strictly (Harris, 1984, and Brown, 2004) under the researcher’s supervision. Additionally,

the qualitative data indicates how EMI students participate in and perceive EMI educational activities combining multimodal techniques. In order to process the data systematically, the analysis of qualitative data involves labeling, coding, and categorizing to clarify the patterns and themes effectively and directly from the collected data (Yin, 2009). The two-sample *t*-test (also called the independent samples *t*-test) and the paired *t*-test are probably the most widely used tests in statistics for the comparison of mean values between two samples. (Xu M; 2017). Therefore, it was used in this research to compare the empirical data between the CG and EG groups in the pre-tests and post-tests of the students.

### Intervention baseline

To improve students' natural engagement and transit more working time to students, the EG group received the intervention with the combination of multimodal instruction in each lesson of the three Units. These experimental lessons were extended to 9 weeks in the second semester of a class in a public primary school to observe the changes of both students and the teacher following the Figure 2 below by the researcher.



**Figure 2. The intervention procedure in EG group**

## FINDINGS

### The impact of multimodal instruction on speaking proficiency in EMI classes

Looking at Table 1, this is the data collected from pre-tests and post-tests between two groups, the experimental group and the control group, using independent-samples *t*-tests. The assessments of two groups utilized the benchmark of speaking criteria (Harris; 1974) to evaluate students' performance in various aspects. To facilitate the analysis and evaluation, the researcher presents the final speaking performance data based on the average of the five elements with the scale of 5 in the speaking assessment.

Before implementing the multimodality in the class, students in both two groups were asked to join the pretest. Whereas the CG group achieved the mean score at around 2.3, the mean of the EG was a bit higher than it, at around 2.36. After the intervention was conducted, the EG group increased the score from 2.36 to 4.17 with the lower SD index compared to the CG group (3.15).

These findings suggest that students benefited from the multimodal intervention. Notably, the CG group's post-test results after the implementation phase of only the EG group also increased because they were also experienced in learning the same topics from their teacher. However, the performance was not as outstanding as the EG group. This demonstrates that multimodal instruction will not impact students' overall speaking results but might contribute to improved performance in combination with the teacher's existing approaches.

The significant difference between the pre-test and post-test of the EG is also shown through the reliable index ( $t=-13.02$ ,  $p<0.001$ , Cohen's  $d=-1.78$ ). From this, these results indicate that the intervention contributed to effective results after three topics extending in the period of 9 weeks with 3 stages of teaching.

| Paired-samples <i>t</i> -tests | Mean     |           | SD       |           | <i>t</i> | <i>p</i> | Effect size<br>(Cohen's <i>d</i> ) |
|--------------------------------|----------|-----------|----------|-----------|----------|----------|------------------------------------|
|                                | Pre-test | Post-test | Pre-test | Post-test |          |          |                                    |
| Control Group (N=40)           | 2.31     | 3.15      | 0.66     | 0.64      | -6.44    | .000*    | -1.02                              |
| Experimental Group (N=43)      | 2.36     | 4.17      | 0.63     | 0.57      | -13.02   | .000*    | -1.78                              |

**Table 1. Within-group pre-test and post-test results**

| Independent-samples <i>t</i> -tests (CG vs EG) | Mean |      | SD   |      | <i>t</i> | <i>p</i> | Effect size<br>(Cohen's <i>d</i> ) |
|------------------------------------------------|------|------|------|------|----------|----------|------------------------------------|
|                                                | CG   | EG   | CG   | EG   |          |          |                                    |
| 1. How to save natural resources?              | 3.02 | 4.05 | 0.70 | 0.60 | 7.24     | .000*    | 1.58                               |
| 2. How to use senses to learn about matter?    | 3.14 | 4.18 | 0.68 | 0.56 | 7.68     | .000*    | 1.66                               |
| 3. How to describe matter around you?          | 3.29 | 4.28 | 0.66 | 0.55 | 7.51     | .000*    | 1.61                               |

**Table 2. Between-group post-test comparisons**

In addition, to consolidate the results more reliably, the researcher also synthesized the index between the CG and EG (Table 2) to observe the important changes taking place after the application of multimodality in each topic. In the first topic of Natural Resources, the EG surpassed the CG ( $t=7.24$ ,  $p<0.001$ , Cohen's  $d=1.58$ ). In the next two topics, the same trend was observed in two groups, particularly through the oral presentation in the topic of Matter and Senses.

In the context of EMI classes, other criteria such as *micro*- and *macro*-skills of speaking are similarly important. In order for the observation to be processed, it was made deliberately by the teaching assistants in two environments in the observation sheets after each presentation of each group to prove the great changes and differences in two groups. Particularly, some sample speeches were noted down (Table 3) below.

|                             | EG                                                                                                                                                                   | CG                                                           |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Ability of producing chunks | EG1: Group 1: To protect natural resources, we should recycle,... save electricity,...and....plant more trees.                                                       | CG1: Group 3: Natural resources are things from..... nature. |
| Stress patterns             | EG1: Group 2: We use trees to make paper and furniture.<br>We <i>also</i> use sunlight to produce solar energy. <i>However</i> , some natural resources are limited. | CG3: Ice is cold and hard.<br>Water is wet and soft.         |

|                                  |                                                                                                                                                                                         |                                                                                                      |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Adequate number of lexical units | EG2: Group 5: <i>Careful obser..vation</i> helps <i>scientists discover</i> new <i>information</i> and <i>solve problems</i> .                                                          | CG2: Group 2: I use my eyes to... see colors. I use my ears to....hear music.                        |
| Fluent speech at different rates | EG3: Group 4: <i>So I think</i> , in everyday life, observing matter <i>also</i> helps us make decisions, <i>such as</i> choosing clothes for the weather or checking if food is fresh. | CG1: We should save.... natural resour...ces. We should not waste.....water or..... e..lec..tricity. |

Table 3. Students’ excerpts in oral presentations

As you can see in table 3 above, students in the EG group were able to produce chunks in the complex sentences in the sample speech of Group 1, whereas students in the CG group were unable to say the whole difficult words such as “natural” or “resources”. In the second row, students in the EG group could utilize the connectors such as “also”, “however”...to emphasize the similar idea or the opposite idea between two sides. However, the CG group only made simple and basic sentences. This observation sheet provided details and facts happening in the class in order that the researcher could confidently state that multimodal application in the class works effectively in adjusting students’ speaking performances, particularly in the EG group.

Students’ responses to multimodal instruction in terms of classroom engagement

In this research, we aim to increase students’ engagement by utilizing a multimodal instruction mainly throughout each session of the topic. Therefore, it is necessary to observe the reactions of the students during the process with their peers to make changes in time to orient students to the target of the lesson. There are three different modalities to diagnose the interaction, including *physiology, behavior, and performance* (Imène Jraidi; 2013), using a dynamic Bayesian network. During the process of cooperating with their partners, the first aspect to consider was the performance of students. Most of them could answer correctly the questions (Appendix C), approximately 78.13%. This percentage suggested that the interaction using the multimodal instruction of the teacher from the first session worked effectively. Along with this result, throughout the observation, there were only 1-2 students reported off-task or non-interacting with their teammate.

| N (i=5) | Correct (1st) | Incorrect (1st) | % Correct (2nd) | % Correct (2nd) |
|---------|---------------|-----------------|-----------------|-----------------|
| 43      | 33,6          | 9,4             | 78,13%          | 96,8%           |

\*i: a number of questions for each student

Table 4. Students’ results in the interview in the session 2 with friends

The proportion above also indicates the fact that students still struggled with some questions, with which they were in the state of being stuck or had the unfavourite interaction. This was shown through their behaviours such as frowning eyes, silence, or even questions... (Table 3). Therefore, a partner’s support or teacher’s adjustments were essential to ensure students’ engagement during the activity. In this process, teachers’ support only occurred when their friends could not address or answer the questions. Several examples of authentic questions are listed below (Table 3) to capture how effective the multimodal instruction was in supporting the explanation without bringing the first language into the class immediately.

| Interviewee’ questions/comments during the interview | Interviewer’s support | Teacher’s support |
|------------------------------------------------------|-----------------------|-------------------|
|------------------------------------------------------|-----------------------|-------------------|



|                                                      |                                                                                                                         |                                                                                                 |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1. What is sense?                                    | Use embodied modes: hand gestures and emotional expression to explain. For example: point to each sense on his/her body | No support                                                                                      |
| 2. How to say...<property>?<br>3. What does it mean? | Use visual aids: like table, TV... (partner points to the item)                                                         | Use disembodied modes: audio from online dictionary                                             |
| 4. Too difficult, I skip                             | Use interpersonal relationships: let's me explain, let's do it together...                                              | Use interpersonal relationships: think carefully and answer it, one mystery gift for your pair. |
| .....                                                |                                                                                                                         |                                                                                                 |

**Table 5. Students' reactions in the pairwork activities**

As can be seen in Table 3, students first showed their stuck interaction through the specific question related to the vocabulary. At this time, peers' support using embodied modes took place to ensure the discussion going on. This support also reinforced the helper's revision of knowledge, playing a role as an instructor to review the knowledge with their friends. In some cases, support from both teacher and peer is necessary to consolidate the understanding more certainly. Whereas the peer used visual aids such as realia or pictures to explain, the teacher could utilize disembodied modes from online sites such as online dictionaries or YouTube videos. Additionally, sometimes students also encountered mental problems, indicating them through their sayings, such as "too difficult, I skip...". At this time, interpersonal relationships were intervened with to boost their motivation and inspiration intrinsically. That also indicated the importance of building up the relationships between peers and peers and teachers.

In addition, the researcher relied on these elements to assess students' engagement at the last lesson of each Unit based on different indicators in Table 4 below. The observer rated each indicator above based on a five-point Likert scale (1-5). The overall engagement score was the mean of eight indicators shown in Table 6.

| Indicator                         | Description                                                                     | Before Implementation Mean | After Implementation Mean | Improvement |
|-----------------------------------|---------------------------------------------------------------------------------|----------------------------|---------------------------|-------------|
| On-task Behavior                  | Students stay focused on learning activities and follow instructions.           | 2.6                        | 4.4                       | 69,2%       |
| <b>Voluntary Participation</b>    | Students actively answer questions, share ideas, and participate in activities. | 2.1                        | 4.2                       | 100%        |
| <b>Student Talking Time</b>       | Students actively use English during pair/group discussions.                    | 2.3                        | 4.3                       | 87%         |
| <b>Interest and Enjoyment</b>     | Students show positive emotions and interest in learning tasks.                 | 2.8                        | 4.5                       | 60.7%       |
| <b>Confidence in Speaking</b>     | Students speak without the fear of making mistakes.                             | 2.4                        | 4.1                       | 70.8%       |
| <b>Task Persistence</b>           | Students keep working with challenging tasks.                                   | 2.7                        | 4.2                       | 55.6%       |
| <b>Use of Learning Strategies</b> | Students apply gestures, vocabulary...to complete the tasks.                    | 2.5                        | 4.0                       | 60%         |

|                    |                                                                          |                |                |       |
|--------------------|--------------------------------------------------------------------------|----------------|----------------|-------|
| Peer Support       | Students help and cooperate positively with classmate during activities. | 2.9            | 4.6            | 58.6% |
| Overall Mean Score |                                                                          | 2.54/5 (50,8%) | 4.29/5 (85,8%) | 68.9% |

**Table 6. Comparison of students’ engagements before and after the implementation**

As can be seen in Table 6, classroom observations revealed a considerable increase in student engagement after the implementation of multimodal instruction. The overall engagement score increased from 50.8% to 85.8% after the intervention. Notably, voluntary participation and student talking time were the two greatest improvements, proving that students were more willing to interact and communicate in English. In addition, two other indicators, which are on-task behavior and confidence in speaking, saw the second most changes, with the improvement percentage around 69.2% and 70.8%, respectively. This presents the fact that multimodal instruction may play an important role in engaging students in the class.

**DISCUSSIONS**

**Multimodal instruction facilitates comprehension and enhances speaking proficiency**

Research on multimodal instruction combining in EMI context to boost speaking proficiency has increasingly concerned by many researchers recently.

Cheng-Ji Lai (2024) highlights the impact of hands-on activities versus worksheet-based learning on fourth-grade students’ English oral communicative competence, focusing specifically on content gains assessed through oral presentations. However, the author emphasised more how much content students could gain instead of improving students’ proficiency and fluency, whereas Vietnamese primary students are lacking this.

Tonnessen (2010) indicated that the use of semiotic signs and kinaesthetics is often ignored by teachers of English in the lower grades, yet they play a fundamental role in the acquisition of listening and speaking skills. Compared to other researchers such as Jewitt (2014) and Lim (2021), they also stated that the literacy ability of a learner is deemed to be skewed without comprehending the use of the visual, aural and kinaesthetic in tandem with the verbal and print text approach in the contemporary world. Most of the researchers did not mention the way to help students apply semiotic signs such as hand gestures, facial expressions or visual aids in their own speaking to support their presentation skills, enhance their inspiring speech and facilitate their listeners’ understanding. More than that, they just focused on how the teachers utilize these modes.

Since the objectives of the study is to find out the effective multimodal instruction to enhance students’ speaking proficiency even in EMI classes, the results of the research were relatively positive.

As can be seen above in this research, the EG group outperformed the CG group in the post-test with the Cohen’s *d* range: 1.58 – 1.61, with the evaluation taking place during the interaction called Dynamic Assessment and oral presentations of each topic in the intervention stage. This accomplishment suggests that multimodal approach brings about the benefits in providing students

clear paths how to not only develop their proficiency but also broaden their knowledge about social contents.

The results are indicated through the steady index in three topics after constituting the mean score of five aspects of speaking skill. The improvement may be attributed by the application of visual, auditory; and most importantly embodied modes such as gestures and facial expressions, voice tone which plays a role as a meaning maker in delivering lessons effectively. It was also stated by several researchers that multimodal instruction also enhances their language proficiency to be fluent speakers and develops their learning performance. (Merchan, Carrasco and Soto (2015)).

During the oral presentations, the EG students were required to synthesise their learning and present their understanding coherently. This task not only reinforced their procedural knowledge but also demanded higher-order thinking skills, such as organisation, synthesis, and articulation of ideas. This process of working in pairs and groups allowed students to discuss, solve problems, and raise ideas by applying their own multimodal instruction and authentic language. Zhussupova et al (2022) suggests that the systematic use of multimodal texts in the classroom contributed to increasing the level of efficiency in mastering the material, stimulating the cognitive aspects of learning, such as the perception and awareness of information; increasing the motivation of students to study; developing teamwork skills; and activating and enhancing students' speaking skills. Similarly, presenting their ideas in front of the class and having discussions in pairs would contribute to the proficiency of speaking skills when they have to master the content to perform in the class.

From the above statement, the findings suggest that learners can be enhanced and improve their speaking proficiency through the combination of multimodal instructions in each lesson based on producing their own creative multimodal modes in peer interaction and also through oral presentation. Moreover, students could use embodied modes, such as gestures and facial expressions, to present their ideas and facilitate the speaking content, which many other researchers did not focus on helping students to master except for speaking proficiency.

### **Multimodal task design promotes student autonomy and engagement**

The study also demonstrates the effectiveness of combining multimodal instruction in different hands-on activities such as peer interaction and oral presentation in improving student engagement. Students can be encouraged to achieve developmental milestones through social and peer interaction (Psomos & Kordaki, 2012). In this research, student talking time and active participation of students increased considerably after the implementation of multimodal instruction. One important variable is task design, and regarding this, Mercer and Dörnyei (2020) distinguish three different levels: (1) physical appeal, (2) activity appeal, and (3) content appeal. With the detailed design of three-stage instructions, students could cooperate more with their peers during each lesson. This helps promote students' autonomy in actively operating the classroom but with the teacher's orientation of the objectives in the first lesson.

Also, students' autonomy should be encouraged in the lessons by the teacher to allow them to choose what they would like to study instead of pushing them to follow unconceptualised steps or arrangements. Teachers could be flexible to adjust the activities based on the teacher's dynamic assessment to orient them to the target language by a combination of various methodologies. In the second step, students discussed mostly with their partners throughout the interview worksheet. At this stage, students actively implemented creatively a multimodal instruction in explaining and

discussing with their peers, whereas the teacher only played the role of an instructor in the class. Similarly, Masinde et al (2023) also stated that learners can be exposed to exercising their creativity and autonomy by viewing the pictures and drawing, thereby producing their own multimodal modes through their own creativity. Therefore, the student's talking time increasing significantly means they engaged significantly in the class.

At the same time, high engagement was all shown clearly throughout the significantly higher percentages of their confidence and on-task behaviour after implementing multimodal instruction (embodied, disembodied, and interpersonal relationships) in the EMI class. Plus, the feedback provided by the teacher and the peers throughout dynamic assessment contributed to fostering the students' engagement by meeting the three fundamental needs postulated by self-determination theory (Ryan & Deci, 2017). By incorporating embodied learning strategies and multimodal task design, educators can create enriched learning experiences that foster deeper understanding and engagement with subject matter.

## **CONCLUSION**

As mentioned in this study, a limited study examined the empirical results of oral presentations embedded with multimodal elements and social content. This study posits some empirical outcomes showing the feasibility and suitability of using dynamic assessment and oral presentation based on multimodal elements for students' speaking performance. This study provides initial data that oral presentation can be utilized as a pedagogical tool for enhancing students' speaking performance. Using the quasi-experimental method supports comparing the authentic teaching flow and its problems in operating classes. The main bottlenecks were underlying mainly the instruments and activities that the teachers mostly used in the class, which were still revolving around teacher-centered and lecture-based deliveries. After the implementation of three main stages in the experimental class, the findings suggested the detailed effective application of the multimodal instruction in the EMI class. Whereas embodied modes appear to support elaborating abstract words in EMI lessons. Therefore, teachers are suggested to combine this aspect more deliberately into the class to convey the lessons more interestingly and effectively. This also provided valuable insights for educators and the curriculum designers in similar contexts.

## **Pedagogical implications and suggestions**

The researcher suggests schools, managers, educators, and teachers develop a well-designed curriculum along with specific lesson plans, supportive instruments, and materials. can help leverage the teachers' confidence in running the class in the time limit and foster students' background knowledge acquisition and language development. A suggested sequence includes 5 minutes for topic instruction with realia, 10 minutes for multimodal instruction, 10 minutes for pair interviews with dynamic assessment, and 10 minutes for oral presentation and feedback. Teachers should carefully balance worksheet-based instruction and oral presentation to consolidate their knowledge in both speaking skills and lexical and grammar outcomes. They are encouraged to flexibly employ various aspects in the multimodal instruction based on the students' interaction and behavioral philosophy. Furthermore, pair and group discussions should provide students with the autonomy to raise opinions and ideas in the pair and group discussions; it provides them with respect, proactive attitudes, and pleasant emotions as they think it is beneficial to them, yet the moderator ensures the outcomes.

## **Limitations and suggested for the future study**

The study's limitations included the slight variations in the surroundings and English proficiency of the two teachers. Teachers' teaching styles, interactions, attitudes, and mindsets may have a minor influence on the differences in results and applications. Another area for improvement is related to the qualitative data analysis, with the evaluation of beginning situations of students in two classes. Therefore, interrater reliability measures could have been adopted to increase the consistency of the analysis from the qualitative data-collection instruments. Moreover, the intervention period was relatively short and only applied to one subject, which was science, so it might be necessary to investigate the effectiveness of multimodal instruction in other EMI subjects and grades to be able to implement it generally and broadly. Since then, other researchers could extend the number of intervention days to prove the reliability and stable development of students' speaking proficiency and engagement. Also, the list of evaluators and teachers' qualifications and other pedagogical factor descriptions needs to be measured by quantitative data to ensure the variations are low and result in more reliable data to conduct the intervention.

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